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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MEE 322 — Past Questions & Materials
by Renance

Course: MEE 322

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 1

Levels: 300

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Paper 1 of 1 — MEE 322 Mechanics of Machines II

Level 300 • Second Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

DEPARTMENT OF MECHANICAL ENGINEERING

Second Semester Examination 2023/2024 Session

COURSE. CODE AND TITLE: MEE 322- MECHANICS OF MACHINES If

TIME ALLOTTED: 2.HOURS

INSTRUCTION: ANSWER QUESTION 1 (COMPULSORY) AND ANY OTHER THREE QUESTION é
QUESTION 1 |

(a) In a table format, highlight the applications/uses, advantages vs.

limitation, schematic/graphical

representation of the following threads: square thread, Acme thread, butt

ress thread and round thread.

(3 Marks)

(b) Calculate the length of the bearing required for a 20 mm diameter shaft supporting a 2.0 kN load, with *

an allowable bearing pressure of 8N/mm². P. _---- eats (4 Marks)

(c) Define any ten (10) of the following terms or systems: (i) profilometry, (ii) oil whirl, (iii) elastohydrodynamic lubrication, (iv) fretting, (v) Stribeck curve, (vi) abrasion (vii) camshaft, (viii) multi-part

bearing, (ix) split of belt, (x) universal coupling, (xi) micro component, (xii) flywheel, (xiii) torque converter,

(xiv) pour point depressant, (xv) polycrystals, (xvi) eccentricity ratio, (xvii) partial bearing (5 Marks)

(d) Complete the statement "Continuous Swinging of a long belt can x,,,,,, while the tight side of the belt can

Cc (d) Highlight the transition between the boundary (thin film) and hydrodynamic (thin) lubrication (3 Marks)

(e) How does 3D printing technology impact the design and manufacture of mechanical parts? (2 Marks)

(f) An open belt drive connects two pulleys 1.2 m and 0.5 m diameter, on parallel shafts 4 m apart. The mass

of the belt is 0.9 kg/m and the maximum tension should not exceed 2000 N. The coefficient of friction is

0.3. The 1.2 m pulley, which is the driver, runs at 200 rpm. Due to belt slip on one of the pulleys, the velocity

of the driven shaft is only 450 rpm. Calculate

the power transmitted, ~

and power lost in friction. What is the efficiency? (7 Marks)

QUESTION 2

(a) The pitch of 50 mm main diameter threaded screw of a screw jack is 12.5 mm. The coefficient of friction between the screw and the nut is 0.13. Determine: (i) the torque required on the screw to raise a load of 25

kN, assuming the load rotate with the screw, and (ii) the efficiency of the machine (9 Marks)

: (b) Discuss the effect of temperature on absolute viscosity of the lubricating oil and gases, (6 Marks)

QUESTION 3

(a) Under what conditions do power or motion is transmitted from one shaft to another with gear drive?

(8 marks)

(b) With the aid of Fig. 3b, elucidate the phenomenon of the transition between the boundary (thin film)

and hydrodynamic (thick film) lubrications (7 Marks)

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Fig. 3a

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QUESTION 4

(a) Enumerate the properties of lubricant additives and outlines the various types of additives and their uses.

(7 marks)

(b) With the aid of Table 4b, compare the areas of application and challenges as they are related rigid and

universal couplings (8 marks)

Types of coupling) Area of application Technical challenges

Rigid coupling

Universal coupling

QUESTION 5

(a) What are the applications of Tribology in Nano-technology and Mechanical Engineering? (8 marks)

(b) An oil film is distributed on a fixed surface. A rectangular plate without mass is placed on the oil film and

It moves horizontally at a constant speed by a distance, s , in the time t . Determine the frictional force that

controls the displacement? assuming the oil is a Newtonian fluid, (10 marks)

Table Q5b. Datasets on the oil film distributed on a fixed surface

Lubrication gap/height (h), mm 1.7mm | Viscosity (η) | 13 mPas ~ ~

| Width of plate 450mm | Distance (s) 0.4m

Length of plate 300 mm | Measured time 2s

L.

QUESTION 6

(a) Highlight factors employed in classifying lubricants by application and lubricants by additives (6 marks)

(b) A body resting on a rough horizontal plane required pull of 180 N inclined at 30° to the plane just to

move it. It was found that a push of 220 N inclined at 30° to the plane just moved the body.

Determine the

weight of the body and the coefficients of friction. (9 marks)

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